

OMAR ABDELHADI

 omarabdelhadi.com

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Education

Carleton University

Sep 2020 – Dec 2024

Bachelor of Computer Science (Honours) - Artificial Intelligence and Machine Learning, GPA: 3.65/4 Ottawa, Ontario

Technical Skills

Languages: Python, Java, C/C++/C#, HTML/CSS, JavaScript, TypeScript, SQL

Tools & Frameworks: Node.js, React, Tailwind, Next.js, Docker, Jenkins, Git, GitHub, Bitbucket, Vite, NumPy, Pandas

Cloud & DevOps: AWS, Azure, CI/CD Pipelines, Kubernetes, Terraform, Jira, Blackduck

Databases: MySQL, MongoDB, Microsoft SQL Server

Operating Systems: Linux, Mac, Windows

Experience

Software Developer

Jan 2024 – Current

3 Canadian Space Division (DND)

Ottawa, Ontario

- Developed **Java**-based back-end solutions using **Maven** and **Spring**, creating **cron-scheduled scripts** and **API endpoints** to automate data ingestion, organize **databases** efficiently, and support front-end data access for analysis.
- Designed and delivered interactive, customizable **graphs, maps, and diagrams** on the **React** front end using **TypeScript**, incorporating analytical calculations to help stakeholders monitor key metrics effectively.
- Deployed applications to **Azure** using **Tanzu** and implemented comprehensive test suites for both back-end services and front-end components, ensuring high code quality and reliability.

Software Engineering Intern

Sep 2022 – Aug 2023

Barracuda Networks LTD

Ottawa, Ontario

- Developed and enhanced **.NET C# microservices** and **REST APIs**, implementing new features and resolving issues to support the functionality of a **RMM** application.
- Built interactive **React** front-end components and integrated them with **backend APIs**, ensuring seamless user interaction and data flow.
- Utilized **AWS** tools to monitor and maintain the state of services, while designing **CI/CD pipelines** to automate code scans, enforce **code quality**, and manage **package licenses** effectively.
- Engineered a **low-level system agent** to interface directly with Windows OS components, using **C#** and **scripting** to gather and process critical **SaaS data**, like hardware, software, and performance metrics, for downstream analysis.

Web Developer Intern

May 2022 – Aug 2022

Getit Local Technologies

Ottawa, Ontario

- Developed and optimized **React Native** mobile applications and **React**-based web interfaces, ensuring seamless user experiences across **iOS, Android, and desktop** platforms.
- Designed **MongoDB** queries to handle large datasets, while analyzing and optimizing query performance each time.
- Implemented **REST APIs** in **Node.js** to enable efficient data exchange between the backend and front-end.

Teacher Assistant

Jun 2021 – Dec 2024

Carleton University

Ottawa, Ontario

- Guided students in programming labs and workshops, providing support in **JavaScript, Python, and Java** for web development courses.
- Led weekly sessions and office hours to assist with debugging and clarify complex concepts, fostering student understanding.
- Collaborated with faculty to refine course materials and enhance the overall learning experience in web technologies.

Projects

Canny Edge Detector | Python

Jan 2024

- Implemented the **Canny Edge Detection** algorithm, utilizing techniques such as **Gaussian blurring**, gradient calculation, and edge tracking by hysteresis to detect edges in images.
- Visualized edge detection results on various images with varying **sigma** and **threshold** values, showcasing potential applications in lane detection technology.

AED Simulation Software | QT C++

Dec 2023

- Developed a prototype for **AED Simulation** to replicate key functions of an Automated External Defibrillator (AED), enhancing user preparedness for cardiac emergencies.
- Designed the system to guide users through heart rhythm analysis, shock administration, and provide real-time assistance during emergencies.